Activity: Session 2.1

Topic: Intelligent Agents: agents and environments, good behavior

Task: Discuss a simple agent function for the vacuum-cleaner world.

1. The Vacuum-Cleaner World

The world has just two locations, square A and square B. Each square may or may not contain dirt. The vacuum agent perceives which square it is in and whether that square is dirty.

Environment	Two squares, A and B; each is either Clean or Dirty
Percepts	[Location, Status], e.g. [A, Dirty] or [B, Clean]
Actions	Left, Right, Suck, NoOp (do nothing)
Sensors	Location sensor and dirt sensor
Actuators	Wheels (to move left or right) and a suction mechanism

2. The Simple Agent Function

An agent function maps every possible percept sequence to an action. One very simple agent function is:

If the current square is dirty, then Suck; otherwise, move to the other square.

Pseudocode (a reflex agent):

```
function REFLEX-VACUUM-AGENT([location, status]) returns an action
    if status = Dirty then return Suck
    else if location = A then return Right
    else if location = B then return Left
```

3. Partial Tabulation of the Agent Function

This agent function depends only on the current percept, so the table can be filled in directly. The first rows are shown below.

[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
...	...
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck

...	...

4. Discussion Questions

1. **Is this agent rational?**
   Ans: Rationality depends on the performance measure, what the agent knows about the environment, the actions it can perform, and its percept sequence so far. Consider a measure that awards one point per clean square per time step over 1000 steps. Under that measure, with a known map and reliable sensors, the agent does well.
2. **When does it become irrational?**

   Ans: If every move costs one point, the agent keeps shuttling between clean squares forever and loses points for no benefit. A better agent would stop (NoOp) once both squares are clean.
3. **What if the environment changes?**
   Ans: If the agent does not know the geography, or squares can become dirty again, the agent should explore, remember what it has seen, or learn from experience.
4. **What if sensors or actuators are unreliable?**
   Ans: A faulty dirt sensor or a Suck action that sometimes fails would require repeating actions or checking the result.
5. **How can the agent be improved?**
   Ans: Add memory (a model of the world), goals (all squares clean), or a utility function that trades off cleaning against energy used.

5. Key Takeaways

- An agent function is a mapping from percept sequences to actions; the agent program is its concrete implementation.
- A rational agent maximizes its expected performance measure, so the measure must be defined carefully.
- Even a simple reflex agent can be rational in one environment and irrational in another.